



DEALER BEST PRACTICE SERIES

Cell-Specific Service Reports

Application Maintenance Site Management **Component Rebuild** Component Life Management Safety MARC Management

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1.0 Introduction

Service reports are critical to effective CRC operations. Service report writing is non-value adding and is typically performed differently by every technician in the shop. Training technicians to write “good” service reports is performed many different ways as well. Since the typical technician is seldom interested in spending time writing, comprehensive/consistent service reports are even harder to get.

Effective service reports should be as brief as possible, while providing all the critical repair-related information. In general, the information is summarized as:

- **Complaint** – Voice of the customer. Symptoms, etc.
- **Cause** – Root cause of failure, including the part and group numbers, etc.
- **Correction** – How was the root cause corrected, and the resulting damage fixed, etc.

Properly written service reports are valuable for several purposes:

- **Customer** - Service-billing justification for additional work performed, operation corrections, maintenance enhancements, etc.
- **Warranty** – “Original” revenue repair clarification for assessing warranty application. Service claim narrative development, and audit trail (proof of repair), etc.
- **Rebuild** – Last repair/rebuild description/clarification for assessing the worn-out core’s rebuild-ability, etc.
- **Caterpillar** – Failure related information for assessing the vehicle’s quality/capability, etc.

Excellent service reports provide all critical information for several users. Given the repetitive nature in many CRC workstations, several dealers have simplified service reporting with special cell-specific reporting forms. Such forms combine the benefits of a service checklist and service report into one document, saving the technician time in writing. This Best Practice will address such service reporting methods and introduce examples of how dealers employ process checklists in their service report forms.

2.0 Best Practice Description

Traditional service reporting simply utilized the familiar orange and white Caterpillar service report. The form includes complaint, cause, and correction areas. Additional notes can be added on the bottom of the form. Limitations of this type of reporting include:

- Technicians often do not complete much of the root cause information with machine-specific identity because sub-component cores arrive with no machine serial number.
- The report does not prompt the technician to include specific repair-related data such as measurements, etc.
- Technicians typically include excess information when it is difficult for them to remember what data is really needed. They spend many hours writing to “cover themselves,” “just in case”. This wasted time expands when there is a lesser repair workload and the technicians need to keep themselves busy.
- Those reviewing service report histories find themselves reading everything reported on a job instead of scanning the form for the information they need. Many pieces of critical

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information are not recorded due to the technician trying to remember too many job-related issues. The report is written like a book with no index.

- Many warranty repair jobs are returned to the CRC for further explanation or data to better explain failures and corrections. Returning jobs is more frequent with non-Caterpillar OEM products due to the low staff experience with these situations and the required product data.
- Some dealers write service history with lead employees or technical writers, due to the difficulty in capturing effective reporting from the technicians. Although these reports may be more readable, they often end up under-reporting critical information due to communication gaps that typically occur when trying to relay highly detailed information. This method is also typically more costly than effective service reporting systems, given the number of employees involved.

In general, traditional service reporting is a very generic process with little guidance to assist in capturing comprehensive and critical information. This system costs dealers:

- Excess labor when technicians over-report.
- Excess administrative efforts when reported information is extensive or disorganized.
- Excess cash flow expense and worsened customer relations when technicians under-report.

An improved service-reporting scheme provides several aids for the CRC technicians and administration staff to more effectively capture service information. Reporting aids support rebuild goals of efficiency and quality by prompting job activities (such as spec lookup). Excess (non-value adding) activity is avoided. Such a system includes:

- Cell-specific service reports to guide the technician that performs a relatively repetitive rebuild process. These reports are created to support sub-component, sub-routine, or sub-process rebuild operations:
 - Brake Groups
 - Cam/crank shafts
 - Cooler Groups
 - Cylinders heads
 - Cylinder Blocks
 - Fuel Systems
 - Hydraulic Control Valves
 - Major component assembly
 - Major component disassembly
 - Major component test
 - Pistons, Rods & Liners
 - Planetary Groups
 - Pumps - Oil
 - Pumps - Water
 - Turbochargers
 - Warranty – New Machine
 - Warranty – New Parts
 - Warranty – Non-Cat

These service reports include some items from the traditional generic Caterpillar reports such as areas for Complaint, Cause, Correction, SIMS, etc. They also include specific questions and or part descriptions in fields that require technician response. These responses are in the form of short comments or checkboxes to show iron status or corrective actions. Such prompts are especially important for Non-Cat warranty reports involving obscure data needs that technicians may not frequently encounter.

- Cell-specific service reports use sequenced prompts and responses that match the rebuild process. This consistency facilitates technician-to-technician training and promotes consistent skill development in the new employee. (See note below)
- Cell specific service reports may be called “process checklists”. These reports typically include rebuild task directives to prompt a technician to perform processes. Certain steps may be required for every rebuild, or may only be required by earlier test results (Ex: if a technician can feel a valve stem wear step with his or her fingernail, then he or she must measure the stem diameter). Tasks may be temporarily added to perform product updates or fixes.

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- Cell-specific service reports may include a response field to record test specifications. This field makes it easy for the technician to record all of the needed Caterpillar specs at one time, much more efficient than revisiting the service manual when each spec is needed. The spec fields can also help prevent errors in memorization.
- The cell-specific service reports may be designed for each type of component and for each CRC work area function. The easiest way to distribute these reports is to simply keep them with the other work order information. The reports can be designed to imitate the dealer's electronic service report template and thus serve as a report entry guide.

Note:

- The reporting aids may not appear as useful to an experienced rebuild technician as they would be to an apprentice or inexperienced technician*. While this is true for experienced technicians who perform the same rebuild operations every day, many experienced technicians move around a CRC completing various rebuild segments. These aids become especially useful in prompting such activities.
- The CRC provides a great environment to train new technicians for future promotion into machine repair work. Typical succession plans include first learning about components in the CRC, then machine systems and diagnostics in the machine repair shop with later opportunities in field service. This type of succession planning requires the CRC to continuously replenish itself with new employees. Using cell-specific service reports with process prompting supports the lesser skilled employees and helps sustain consistent quality in key rebuild operations.

*For the purposes of this Best Practice, a rebuild technician ("apprentice") is lesser skilled and would not have the broad experience of a traditional service technician ("journeyman"). The rebuild technician, in general, should be in the grooming stages to move into a diagnostic/repair technician position through a succession plan. This is an important plan since most dealers develop their own technicians and must do so as effectively as possible.

3.0 Implementation Steps

1. Document/find the as-is service report methods and materials in the CRC rebuild areas.
2. Define future rebuild reporting requirements and processes based on expected product type, volume, size, etc.
 - a. Plan for typical rebuild volumes as well as expected repetitive nature of rebuilds.
 - b. Investigate specialized reporting media and gathering/storage/retrieval equipment (template formats, electronic documents, server storage/distribution systems, on-floor terminals, etc.).
 - c. Use Cat Service Operations Dev. and Information Systems expertise as needed.
 - d. Review other dealer systems and results as needed.
 - e. Focus on the typical reporting - 80-90% of reporting/storing/retrieving circumstances.
 - f. Add-in flexible capabilities (such as special reporting templates for product improvements) for minority requirements.
3. Apply improved reporting strategy to operations and shop areas.
 - a. Develop report/checklist templates.
 - b. Consider product updates in the reporting aids.

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- c. Consider reporting enhancement replication outside of CRC (especially R&I functions).
- d. Include shop technician teams in the develop review/application stages.
- e. Include the Standard Job's team in the development process.
- f. Use Cat Service Operations Dev. and Information Systems expertise as needed.
4. Develop/document the new reporting strategy/procedure, with new roles/responsibilities as needed.
5. Adjust
 - a. Repair/rebuild time requirement targets as needed (capture resultant efficiencies).
 - b. Service labor charge-out rates to reflect time and material repair efficiency gains.
 - c. Administrative duties/tasks to reflect new system requirements and efficiency gains.
6. Present strategy to shop employees.
7. Implement reporting processes, templates, and system changes into the shop/office areas.
 - a. This may occur in phases (as needed).
8. Train employees/customers to use the new procedures. Follow-up/enforce immediately.
9. Eliminate previous reporting practices/systems to eliminate old habits.
10. Review process/template performance once established.
11. Establish and implement adjustments as needed.

4.0 Benefits

- Increased capacity - Fewer returned warranty work orders reduces task interruptions for staff - more time available (with less administrative rework) to plan/organize.
- Reduced cost – Fewer returned warranty/policy work orders allows service/warranty staff to focus on current issues to reduce accounts-receivable associated with outstanding claims. Comprehensive reporting justifies revenue to customers, reducing policy allowance expense. Data organization (in tables, not paragraphs) enables faster operations/warranty processing.
- Increase efficiencies - Service technicians are prompted on core reporting requirements, reducing associated labor. Rebuild target times drop, raising standard (flat-rate) job profits.
- Increased quality – Rebuild processes are prompted to be more consistent/accurate, increasing quality, especially with less experienced employees. Training is conducted more consistently and focuses on critical tasks and product updates.
- Improved Operator/Technician Safety – Properly trained technicians perform tasks in a safer manner, reducing overall risk exposure.

5.0 Resources Required

- Equipment Requirement Gaps Vary
 - o Most dealers already have computer and electronic template capability.
 - o Report forms are typically custom-designed and require wide participation for effective buy-in.
- Facility
 - o Technicians need aisle-accessible PC terminal space to retrieve specs and submit electronic reporting (if required). Difficult/expensive for staff to effectively perform this function.
- Support Equipment - Costs vary greatly per region
 - o Technicians need on-floor terminals to effectively retrieve spec's and submit electronic reporting (if required). Difficult/expensive for staff to effectively perform.

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- People
 - o Establish current reporting concerns with team.
 - o Establish process, systems, and equipment improvements through team.
- Training
 - o Define, implement, and enforce the new process, systems, and duties with the production team.

6.0 Supporting Attachments / References

- Component Rebuild Service Operations Development & Assessment (SODA)
- Sample Service Reports (see attachments in this PDF file)
 - o Cylinder Block Inspection & Reconditioning Report
 - o Fuel System Test Sheet
 - o Connecting Rod Checkoff Sheet
 - o 3400 Series Engines – Quality Control Check List

Note: Report form examples are actual dealer examples and do not represent Caterpillar-development.

7.0 Related Best Practices

None.

8.0 Acknowledgements

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